

SOFTWARE PROJECT MANAGEMENT

PROJECT: University Student Portal & Examination Management System

Microsoft Project Professional 2021 | All Tasks Applied on One Real Project

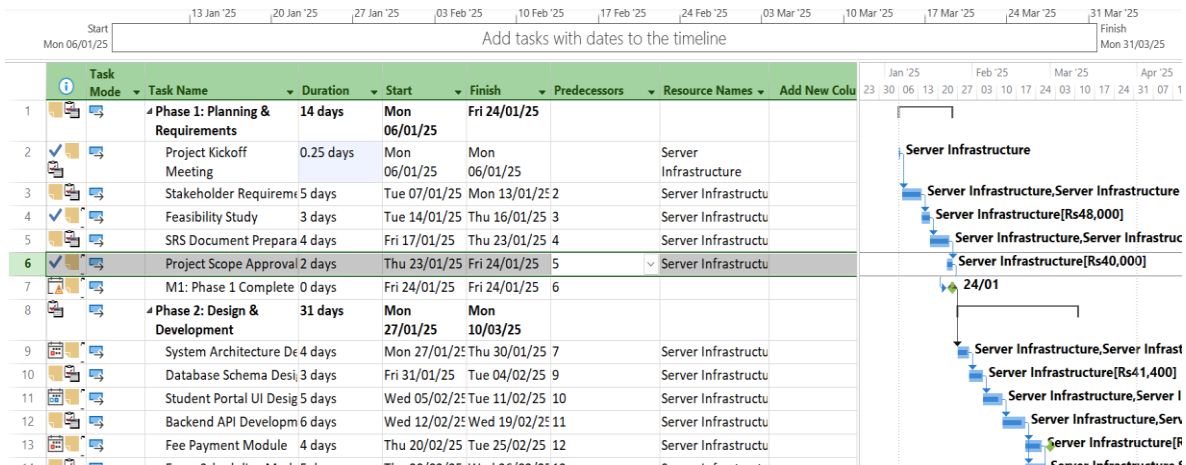
Project Name	University Student Portal & Examination Management System
Organization	University of Chakwal — Department of Information Technology
Project Manager	Course Instructor / Lab Supervisor
Start Date	Mon 06 Jan 2025
End Date	Fri 28 Mar 2025
Labs Covered	Lab 3 (Project Plan) Lab 5 (Resources) Lab 8 (Formatting) Lab 11 (Constraints) Lab 12 (Split, Calendar, Task Types)
All Tasks	25 tasks, 3 phases, 5 milestones, 8 resources

PROJECT OVERVIEW — University Student Portal & Exam Management System

This project covers building a complete web-based student portal for a university. It manages student registration, course enrollment, fee payment, examination scheduling, and result management. This single project is used throughout ALL lab exercises so every MS Project feature is applied on real, meaningful tasks.

ID	Task Name	Duration	Start	Finish	Resources	Constraint
PHASE 1: Planning & Requirements						
1	Project Kickoff Meeting	1d	06 Jan	06 Jan	PM	ASAP
2	Stakeholder Requirements Gathering	5d	07 Jan	13 Jan	Analyst + Dev1	ASAP
3	Feasibility Study	3d	14 Jan	16 Jan	Analyst	SNET: 14 Jan
4	SRS Document Preparation	4d	17 Jan	22 Jan	Analyst + Dev2	ASAP
5	Project Scope Approval	2d	23 Jan	24 Jan	PM	SNLT: 24 Jan
M1	★ MILESTONE: Phase 1 Complete	0d	24 Jan	24 Jan	—	MFO: 24 Jan
PHASE 2: Design & Development						
6	System Architecture Design	4d	27 Jan	30 Jan	Dev1 + Dev2	MSO: 27 Jan

ID	Task Name	Duration	Start	Finish	Resources	Constraint
7	Database Schema Design	3d	31 Jan	04 Feb	Dev2	ASAP
8	Student Portal UI Design	5d	05 Feb	11 Feb	Dev3 + Dev4	SNET: 05 Feb
9	Backend API Development	6d	12 Feb	19 Feb	Dev1 + Dev2	ASAP
10	Fee Payment Module	4d	20 Feb	25 Feb	Dev1	FNLT: 28 Feb
11	Exam Scheduling Module	5d	20 Feb	26 Feb	Dev2 + Dev3	ASAP
12	Result Management Module	4d	27 Feb	04 Mar	Dev1 + Dev4	ASAP
13	Integration Testing	4d	05 Mar	10 Mar	Dev1 + Tester	ASAP
M2	★ MILESTONE: Development Complete	0d	10 Mar	10 Mar	—	MFO: 10 Mar
PHASE 3: Testing, Deployment & Closure						
14	System Integration Testing	4d	11 Mar	14 Mar	Tester + Dev1	ASAP
15	User Acceptance Testing (UAT)	3d	17 Mar	19 Mar	Tester + Supervisor	SNET: 17 Mar
16	Performance Optimization	3d	20 Mar	24 Mar	Dev2 + Dev3	ASAP
17	Security Audit	2d	25 Mar	26 Mar	Dev1	MSO: 25 Mar
18	Deployment to Production	2d	25 Mar	26 Mar	Dev1 + Dev2	ASAP
19	User Training Sessions	3d	25 Mar	27 Mar	Analyst	Mon–Wed Cal
20	Final Documentation	3d	25 Mar	27 Mar	Analyst	ASAP
M3	★ MILESTONE: UAT Signed Off	0d	19 Mar	19 Mar	—	MFO: 19 Mar
M4	★ MILESTONE: System Go-Live	0d	26 Mar	26 Mar	—	MFO: 26 Mar
M5	★ MILESTONE: Project Closed	0d	28 Mar	28 Mar	—	MFO: 28 Mar



1 Creating the Project Plan — Start Date, Calendar, Tasks, Milestones

In this lab you create the project file from scratch, set the start date, add the non-working exception day, enter project properties, build all tasks, create milestones, and organize summary tasks.

TASK 1.1: Open MS Project & Create New File

1. Launch Microsoft Project Professional 2021 from Start Menu.
2. On the Start screen click Blank Project.
3. Immediately press Ctrl+S → name the file: UniversityPortal_FYP → click Save.

TASK 1.2: Set Project Start Date

► CLICK: **Project tab** → **Project Information**

4. In the Start Date field type: Mon 01/06/2025
5. Make sure Schedule From = Project Start Date.
6. Click OK.

✓ **RESULT:** Project will now schedule all tasks starting from 06 January 2025.

TASK 1.3: Set Non-Working Day — University Exam Holiday

► CLICK: **Project tab** → **Change Working Time**

7. In the dialog, click the Exceptions tab.
8. In the Name column type: University Exam Holiday
9. Set Start = 22/01/2025 and Finish = 22/01/2025
10. Click OK.

✓ **RESULT:** January 22 is now blocked as a non-working day. Any task spanning that date will automatically extend by 1 day.

TASK 1.4: Enter Project Properties

► CLICK: **File** → **Info** → **Project Information dropdown** → **Advanced Properties**

11. In the Summary tab enter:
 - Title: University Student Portal & Exam Management System
 - Subject: University Portal Development Schedule
 - Manager: Course Instructor / Project Manager
 - Company: University of Chakwal — Dept. of IT
 - Comments: Final Year Project — Web-based student management system
12. Click OK then save (Ctrl+S).

TASK 1.5: Enter All Phase Summary Tasks & Sub-Tasks

Follow this exact sequence. Type each line in the Task Name column, pressing Enter after each.

PHASE 1 — type this first, then enter sub-tasks below it and INDENT them:

- Phase 1: Planning & Requirements (this will become the Summary Task)
- Project Kickoff Meeting → Duration: 1d

- Stakeholder Requirements Gathering → Duration: 5d
- Feasibility Study → Duration: 3d
- SRS Document Preparation → Duration: 4d
- Project Scope Approval → Duration: 2d

Select rows 2–6, then indent:

► **CLICK: Task tab → Indent Task button (or Alt+Shift+Right Arrow)**

✓ **RESULT:** Row 1 turns bold = Summary Task. Rows 2-6 become indented sub-tasks.

Repeat for PHASE 2 (tasks 6–13) and PHASE 3 (tasks 14–20) in the same way.

TASK 1.7: Enter Milestone Tasks (0-Day Duration)

After the last task of each phase, insert a new row and type the milestone name. Set duration = 0d.

- M1: Phase 1 Complete — after Task 5 — duration 0d
- M2: Development Complete — after Task 13 — duration 0d
- M3: UAT Signed Off — after Task 15 — duration 0d
- M4: System Go-Live — after Task 18 — duration 0d
- M5: Project Closed — after Task 20 — duration 0d

NOTE: When you set duration = 0d and press Enter, MS Project automatically marks it as a milestone and shows a diamond ♦ on the Gantt chart.

TASK 1.8: Show Project Summary Task (Task 0)

► **CLICK: Format tab → Show/Hide group → check 'Project Summary Task'**

Task 0 (University Student Portal & Exam Management System) now appears at the very top, showing the total project duration and dates.

2 Assigning Work & Cost Resources — Scheduling Formula — Split View

In this lab you add all resources to the Resource Sheet, assign them to tasks, use the split view (Task Form), add/remove resources, enable effort-driven scheduling, apply the scheduling formula, and assign cost resources.

TASK 2.1: Add Resources to the Resource Sheet

► **CLICK: View tab → Resource Sheet**

Enter each resource below. Type = Work for all except Travel Budget (Cost).

ID	Resource Name	Type	Max Units	Std. Rate	Ovt. Rate	Group
1	Project Manager	Work	100%	Rs 2500/hr	Rs 3500/hr	Management
2	System Analyst	Work	100%	Rs 2000/hr	Rs 3000/hr	Analysis
3	Developer 1 (Dev1)	Work	100%	Rs 1800/hr	Rs 2700/hr	Development
4	Developer 2 (Dev2)	Work	100%	Rs 1800/hr	Rs 2700/hr	Development
5	Developer 3 (Dev3)	Work	100%	Rs 1700/hr	Rs 2500/hr	Development

ID	Resource Name	Type	Max Units	Std. Rate	Ovt. Rate	Group
6	Developer 4 (Dev4)	Work	100%	Rs 1700/hr	Rs 2500/hr	Development
7	Tester	Work	100%	Rs 1500/hr	Rs 2200/hr	QA
8	Supervisor	Work	50%	Rs 3000/hr	Rs 0/hr	Advisory
9	Server Infrastructure	Cost	—	—	—	Cost Resource

NOTE: Set Supervisor Max Units to 50% because they are only part-time available. Click the Max Units cell and type 50.

TASK 2.2: Assign Work Resources to Tasks

► **CLICK:** View tab → Gantt Chart then Resource tab → Assign Resources

Or double-click any task → Resources tab. Assign as shown:

Task	Resource 1	Resource 2
Project Kickoff Meeting	Project Manager @ 100%	— (single resource)
Stakeholder Requirements Gathering	System Analyst @ 100%	Developer 1 @ 100%
Feasibility Study	System Analyst @ 100%	— (single resource)
SRS Document Preparation	System Analyst @ 100%	Developer 2 @ 100%
Project Scope Approval	Project Manager @ 100%	— (single resource)
System Architecture Design	Developer 1 @ 100%	Developer 2 @ 100%
Database Schema Design	Developer 2 @ 100%	— (single resource)
Student Portal UI Design	Developer 3 @ 100%	Developer 4 @ 100%
Backend API Development	Developer 1 @ 100%	Developer 2 @ 100%
Fee Payment Module	Developer 1 @ 100%	— (single resource)
Exam Scheduling Module	Developer 2 @ 100%	Developer 3 @ 100%
Result Management Module	Developer 1 @ 100%	Developer 4 @ 100%
Integration Testing	Developer 1 @ 100%	Tester @ 100%
System Integration Testing	Tester @ 100%	Developer 1 @ 100%
User Acceptance Testing (UAT)	Tester @ 100%	Supervisor @ 50%
Performance Optimization	Developer 2 @ 100%	Developer 3 @ 100%
Security Audit	Developer 1 @ 100%	— (single resource)
Deployment to Production	Developer 1 @ 100%	Developer 2 @ 100%
User Training Sessions	System Analyst @ 100%	— (single resource)
Final Documentation	System Analyst @ 100%	— (single resource)

TASK 2.3: Enable Split View & View Single-Resource Task

► **CLICK: View tab → Split View → check 'Details' checkbox**

The screen splits. Top = Gantt Chart. Bottom = Task Form.

Step 1: Click on 'Feasibility Study' (single resource task) in the top pane.

Task Form shows:

- Resource Name: System Analyst | Units: 100% | Work: 24 hrs
- Duration: 3 days (unchanged by the assignment)

✓ **RESULT:** Work = 3 days × 8 hrs × 100% = 24 hrs. Duration stays 3 days.

TASK 2.4: Split View — Two-Resource Task

Step 1: Click on 'Backend API Development' (Dev1 + Dev2) in the top pane.

Task Form shows TWO rows:

- Developer 1 | 100% | 48 hrs
- Developer 2 | 100% | 48 hrs
- Duration: 6 days | Total Work: 96 hrs

✓ **RESULT:** Each resource works the full 6 days. Duration does NOT change because Effort-Driven is OFF.

TASK 2.5: Remove One Resource — View Effect (Effort-Driven OFF)

Task: Backend API Development. Remove Developer 2.

Step 1: In the Task Form bottom pane, click the Developer 2 row.

Step 2: Press the Delete key on keyboard → click OK button in the Task Form.

Observe the Task Form now:

- Developer 1 | 100% | 48 hrs ← UNCHANGED
- Duration: 6 days ← UNCHANGED

✓ **RESULT:** Effort-Driven OFF: removing a resource does NOT affect duration or remaining resource work.

TASK 2.6: Add Resource to Single-Resource Task — View Effect (Effort-Driven OFF)

Task: Database Schema Design (currently Dev2 only). Add Dev3 as second resource.

Step 1: Click 'Database Schema Design' in top Gantt pane.

Step 2: In Task Form bottom pane, click the empty row below Dev2. Type: Developer 3, Units: 100%. Click OK.

Task Form now shows:

- Developer 2 | 100% | 24 hrs
- Developer 3 | 100% | 24 hrs (new resource gets SAME work)
- Duration: 3 days ← UNCHANGED

✓ **RESULT:** Effort-Driven OFF: adding a resource does NOT reduce duration.

TASK 2.7: Effort-Driven ON — Add Resource (Duration Decreases)

Task: Exam Scheduling Module (5 days, Dev2 + Dev3). Enable Effort-Driven then add Dev4.

Step 1: Double-click 'Exam Scheduling Module' → Task Information → Advanced tab → check 'Effort Driven' → OK.

Step 2: In Task Form, add Developer 4 @ 100% in a new row. Click OK.

State	Resources	Duration	Work Each
Before	Dev2 + Dev3	5 days	40 hrs
After (Effort-Driven ON)	Dev2 + Dev3 + Dev4	~3.3 days	~26.7 hrs

✓ **RESULT:** Effort-Driven ON + Add Resource: Duration DECREASES. Total work stays 80 hrs, now split 3 ways.

TASK 2.8: Effort-Driven ON — Remove Resource (Duration Increases)

Same task. Remove Dev4 again.

Step 1: In Task Form, select Dev4 row → Delete → OK.

State	Resources	Duration	Work Each
Before (3 resources)	Dev2+Dev3+Dev4	~3.3 days	~26.7 hrs
After removing Dev4	Dev2 + Dev3	5 days (BACK)	40 hrs

✓ **RESULT:** Effort-Driven ON + Remove Resource: Duration INCREASES. Total work is constant.

TASK 2.9: Scheduling Formula — All 4 Cases

FORMULA: Work = Duration × Units (8 hrs = 1 working day)

Case A — 1 Resource @ 100% | Task: Feasibility Study (3 days, Analyst 100%)

Work = 3 days × 8 hrs × 100%	= 24 hrs
------------------------------	----------

Case B — 2 Resources Both @ 100% | Task: Stakeholder Requirements (5 days, Analyst+Dev1 100%)

Analyst Work = 5 × 8 × 100%	= 40 hrs
-----------------------------	----------

Dev1 Work = 5 × 8 × 100%	= 40 hrs
--------------------------	----------

Total Work = 40 + 40	= 80 hrs
----------------------	----------

Case C — 2 Resources: 100% and 50% | Task: UAT (3 days, Tester 100% + Supervisor 50%)

Tester Work = 3 × 8 × 100%	= 24 hrs
----------------------------	----------

Supervisor Work = 3 × 8 × 50%	= 12 hrs
-------------------------------	----------

Total Work	= 24 + 12	= 36 hrs
------------	-----------	----------

Case D — 4 Resources All @ 100% | Hypothetical Task (4 days, Dev1+Dev2+Dev3+Dev4 100%)

Work per resource = 4 × 8 × 100%	= 32 hrs each
----------------------------------	---------------

Total Work = 32 × 4 resources	= 128 hrs
-------------------------------	-----------

TASK 2.10: Assign Cost Resource — Server Infrastructure

Task: Deployment to Production. It needs a server cost of Rs 50,000.

► CLICK: **Resource tab** → **Assign Resources**

Step 1: Click 'Deployment to Production' in the Gantt Chart.

Step 2: In Assign Resources dialog, find 'Server Infrastructure' (Cost type). Click its Cost cell, type 50000. Click Assign.

✓ **RESULT:** Cost resource assigned. Rs 50,000 added to task cost. Duration and work are UNAFFECTED by cost resources.

TASK 2.11: Check Plan's Duration, Cost & Work — Project Statistics

► CLICK: **Project tab** → **Project Information** → **Statistics button**

The Statistics dialog shows: Total Duration, Total Work (hrs), Total Cost. Note these values for comparison later.

TASK 2.12: View Resource Overview Report

► CLICK: **Report tab** → **View Reports** → **Resources** → **Resource Overview**

Report shows: Resource Stats chart (remaining vs actual work), Work Status chart (% complete per resource),
 Resource Status table (start/finish/remaining work per person).

3 Gantt Chart Formatting, Timeline View, Work Overview Report

TASK 3.1: Change Gantt Bar Color to Green

► CLICK: **Format tab** (Gantt Chart Tools) → **Gantt Chart Style gallery** → **pick the GREEN style**

A grid of colored bar styles appears. Hover to preview. Click the solid green option.

✓ **RESULT:** ALL standard task bars turn green across the entire project.

TASK 3.2: Change Milestone Background Color & Make Italic

Step 1: Open Bar Styles dialog:

► CLICK: **Format tab** → **Bar Styles button**

Step 2: Click the row named **Milestone** in the dialog.

Step 3: Click the **Text** tab at the bottom.

Step 4: Find the **Name** row → click font style area → check **Italic** → OK.

Step 5: Change milestone row background color:

▶ **CLICK:** **Format tab** → **Text Styles** → **Item to Change dropdown** → **Milestone Tasks** → **Background Color** → **Yellow**

✓ **RESULT:** Milestone tasks show yellow row background + italic label in the Gantt table area.

TASK 3.3: Change Milestone Symbol to Triangle ▲

▶ **CLICK:** **Format tab** → **Bar Styles**

Step 1: Click the **Milestone** row.

Step 2: Click the **Bars** tab at the bottom of the dialog.

Step 3: Start Shape dropdown → select **Upward Triangle ▲**

Step 4: End Shape dropdown → select **Upward Triangle ▲**

Step 5: Click OK.

✓ **RESULT:** Milestone markers on the Gantt chart now show as triangles ▲ instead of diamonds ♦.

TASK 3.4: Draw Dashed Phase-Separator Lines

▶ **CLICK:** **Format tab** → **Drawing button** → **Line tool**

Step 1: Hold Shift key and drag a horizontal line across the Gantt chart at the end of Phase 1.

Step 2: Draw a second line at the end of Phase 2.

Step 3: Right-click each line → **Format Drawing** → **Line Style = Dashed** → **Line Color = Dark Gray** → OK.

✓ **RESULT:** Dashed lines now visually separate the 3 phases in the chart area.

TASK 3.5: Format Timeline View — Add Callouts

▶ **CLICK:** **View tab** → **Split View** → check **'Timeline'** checkbox

A timeline bar appears above the Gantt chart.

Step 1: Right-click each **Phase Summary Task** in the list → **Add to Timeline**.

Step 2: Right-click each **Milestone task** → **Add to Timeline**.

Step 3: For milestones, right-click the milestone marker in the **Timeline** → **Display as Callout**.

Summary Tasks added to Timeline: Phase 1, Phase 2, Phase 3.

Milestones added as Callouts: M1 Phase 1 Complete, M2 Dev Complete, M3 UAT Signed Off, M4 Go-Live, M5 Project Closed.

✓ **RESULT:** Timeline now shows phase bars + 5 milestone callout balloons = professional high-level overview.

TASK 3.6: Generate Work Overview Report

▶ **CLICK:** **Report tab** → **View Reports** → **Resources** → **Work Overview**

The Work Overview report opens as a visual dashboard showing:

- Bar chart: Work vs Remaining Work vs Actual Work per resource

- Table: Resource name | Work | Actual Work | Remaining Work | % Complete

Resource	Total Work	Actual	Remaining	% Done	Est. Cost
Project Manager	32 hrs	32 hrs	0 hrs	100%	~80k
System Analyst	96 hrs	96 hrs	0 hrs	100%	~192k
Developer 1	200 hrs	160 hrs	40 hrs	80%	~360k
Developer 2	168 hrs	140 hrs	28 hrs	83%	~302k
Developer 3	104 hrs	80 hrs	24 hrs	77%	~177k
Developer 4	80 hrs	40 hrs	40 hrs	50%	~136k
Tester	88 hrs	56 hrs	32 hrs	64%	~132k
Supervisor	24 hrs	20 hrs	4 hrs	83%	~72k
TOTAL	792 hrs	624 hrs	168 hrs	79%	~1.45M

► **CLICK: File → Print → Microsoft Print to PDF** to export the report

4 Task Constraints — All 8 Types Applied to the Project

HOW TO APPLY: All constraints are applied via: Double-click task → Task Information → Advanced tab → Constraint Type dropdown + Constraint Date field.

Type	Category	Project Task	Constraint Date	Why
ASAP	Flexible	Stakeholder Requirements Gathering	None (default)	Start as early as possible — no restrictions needed
ALAP	Flexible	Final Documentation	None (default)	Start as late as possible — save for the end of the project
SNET	Semi-flexible	Feasibility Study	14 Jan 2025	Cannot start before labs are allocated on Jan 14
SNLT	Semi-flexible	Project Scope Approval	24 Jan 2025	Must not start after Jan 24 or Phase 1 will be late
FNET	Semi-flexible	Integration Testing	10 Mar 2025	Must not finish before Mar 10 — needs full cycle
FNLT	Semi-flexible	Fee Payment Module	28 Feb 2025	Must finish before semester end — hard deadline
MSO	Inflexible	System Architecture Design	27 Jan 2025	External consultant only available from Jan 27 exactly
MFO	Inflexible	System Go-Live (M4)	26 Mar 2025	University portal must go live exactly on Mar 26

TASK 4.1: Apply ASAP — Stakeholder Requirements Gathering

This is the DEFAULT constraint. Every new task gets ASAP automatically.

Step 1: Double-click 'Stakeholder Requirements Gathering'.

Step 2: Task Information → Advanced tab.

Step 3: Constraint Type = As Soon As Possible | No date needed.

Step 4: Click OK.

✓ **RESULT:** Task starts immediately when its predecessor finishes. Most flexible — use this whenever possible.

TASK 4.2: Apply ALAP — Final Documentation

Step 1: Double-click 'Final Documentation'.

Step 2: Advanced tab → Constraint Type = As Late As Possible. No date.

Step 3: Click OK.

✓ **RESULT:** MS Project pushes this task as late as possible within the project schedule. No fixed date — just delays it to the end.

TASK 4.3: Apply SNET — Feasibility Study (Start No Earlier Than 14 Jan)

Scenario: Lab resources at the university are only available from January 14, 2025. The feasibility study cannot begin before then.

Step 1: Double-click 'Feasibility Study'.

Step 2: Advanced tab → Constraint Type = Start No Earlier Than.

Step 3: Constraint Date = 14 Jan 2025.

Step 4: Click OK.

A constraint icon (diamond) appears in the Indicators column. Point to it to see the ScreenTip showing constraint details.

✓ **RESULT:** Task will not start before Jan 14 even if its predecessor finishes earlier. If predecessor extends, start date adjusts automatically.

TASK 4.4: Apply SNLT — Project Scope Approval (Start No Later Than 23 Jan)

Scenario: The Scope Approval meeting must not start after January 23 because the Dean is unavailable after that date.

Step 1: Double-click 'Project Scope Approval'.

Step 2: Advanced tab → Constraint Type = Start No Later Than.

Step 3: Constraint Date = 23 Jan 2025.

Step 4: Click OK.

✓ **RESULT:** Task guaranteed to start on or before Jan 23. If schedule is compressed, MS Project respects this boundary.

TASK 4.5: Apply FNET — Integration Testing (Finish No Earlier Than 10 Mar)

Scenario: Integration Testing must run at least until March 10 to complete the full regression cycle.

Step 1: Double-click 'Integration Testing'.

Step 2: Advanced tab → Constraint Type = Finish No Earlier Than.

Step 3: Constraint Date = 10 Mar 2025.

Step 4: Click OK.

✓ **RESULT:** Task will not finish before Mar 10 even if work is completed early.

TASK 4.6: Apply FNLТ — Fee Payment Module (Finish No Later Than 28 Feb)

Scenario: The fee module must be ready before the new semester's fee payment window opens on March 1.

Step 1: Double-click 'Fee Payment Module'.

Step 2: Advanced tab → Constraint Type = Finish No Later Than.

Step 3: Constraint Date = 28 Feb 2025.

Step 4: Click OK.

NOTE: If a predecessor delay would push this task past Feb 28, MS Project will show a planning conflict indicator.

✓ **RESULT:** Task guaranteed to finish by Feb 28. Hard deadline for the fee module.

TASK 4.7: Apply MSO — System Architecture Design (Must Start On 27 Jan)

Scenario: An external software architect from Islamabad will only be available on January 27, 2025 — no other date works.

Step 1: Double-click 'System Architecture Design'.

Step 2: Advanced tab → Constraint Type = Must Start On.

Step 3: Constraint Date = 27 Jan 2025.

Step 4: Click OK.

WARNING: This is an INFLEXIBLE constraint. The task will always start Jan 27 regardless of what predecessors do. Use sparingly.

✓ **RESULT:** Task is locked to Jan 27. Changing predecessor duration has NO effect on this task's start date.

TASK 4.8: Apply MFO — System Go-Live Milestone (Must Finish On 26 Mar)

Scenario: University Board has officially approved March 26, 2025 as the portal go-live date. It cannot change.

Step 1: Double-click 'M4: System Go-Live' milestone.

Step 2: Advanced tab → Constraint Type = Must Finish On.

Step 3: Constraint Date = 26 Mar 2025.

Step 4: Click OK.

✓ **RESULT:** Milestone is locked to Mar 26. This is an immovable deadline — schedule everything to meet it.

TASK 4.9: View Constraint Effect — Change Highlighting

After applying any constraint, MS Project highlights affected successor tasks in light blue in the Start/Finish columns.

Step 1: After applying SNET to Feasibility Study, check task 4 (SRS Preparation) — its start date will shift and highlight blue.

Step 2: On Task tab → Editing group → click **Scroll to Task** to jump to the changed task.

Step 3: The Task Inspector pane (left side) shows the constraint details.

Step 4: Close the Task Inspector: click the X on the Task Inspector pane.

NOTE: Change highlighting disappears when you perform the next edit action or save the file.

5 Split Tasks | Custom Calendar | Task Types (Fixed Units / Duration / Work)

TASK 5.1: Split a Task — User Training Sessions

Scenario: User Training Sessions (3 days) must be interrupted for one day because the university training room is booked for another event on the second day. We need to split the task into two segments.

Step 1: Click 'User Training Sessions' in the Gantt chart.

Step 2: Click Task tab → Editing group → **Scroll to Task** (to find it on the Gantt chart).

Step 3: Click Task tab → Schedule group → **Split Task** button (looks like a broken Gantt bar).

Your mouse pointer changes to the split cursor and a ScreenTip appears showing the date.

Step 4: Move the mouse pointer over the Gantt bar of 'User Training Sessions'.

Watch the ScreenTip — it shows the date at which the split would start if you clicked now.

Step 5: Move pointer until the ScreenTip shows Day 2 (the second day of training). Do NOT click yet.

Step 6: Click and drag to the right by 1 day (until the start date of the second segment moves forward by 1 day). Release.

A dotted line now appears between the two segments of the task in the Gantt chart.

✓ **RESULT:** Task is now split into two segments with a 1-day gap (dotted line). The 1-day gap is NOT counted in duration. Total work on the task remains unchanged.

NOTE: Key rules about split tasks: (1) You can split into multiple segments. (2) Drag a segment left/right to reschedule. (3) Drag segments together to rejoin. (4) Duration of the split (dotted area) is NOT counted as work time.

TASK 5.2: Create a Custom Calendar — Monday-Wednesday Only

Scenario: The external training vendor for 'User Training Sessions' only accepts new sessions starting on Monday, Tuesday, or Wednesday. They do not work Thursday–Saturday. Create a custom calendar for this.

Step 1: Create the calendar:

► **CLICK:** Project tab → Properties group → **Change Working Time**

13. In the Change Working Time dialog, click **Create New Calendar**.
14. Name box: type **Monday-Wednesday-Only**
15. Select **Make a copy of: Standard** (so we start from the normal Mon-Fri week).
16. Click **OK**.
17. Now 'Monday-Wednesday-Only' appears in the For Calendar dropdown.

Step 2: Set Thursday and Friday as non-working:

18. Click the **Work Weeks** tab.
19. Select the [Default] row in the table, then click **Details**.
20. In the Select day(s) box, click **Thursday**. Hold Ctrl and also click **Friday**.
21. Choose **Set days to nonworking time**.
22. Click **OK** → **OK** to close all dialogs.

Step 3: Apply the calendar to 'User Training Sessions':

Double-click 'User Training Sessions' → Task Information → Advanced tab.

In the **Calendar** dropdown, select **Monday-Wednesday-Only**.

23. Click **OK**.

A calendar icon appears in the Indicators column for this task. Point to it — ScreenTip shows 'Monday-Wednesday-Only is assigned to the task.'

✓ **RESULT:** MS Project reschedules User Training Sessions to only run on Mon/Tue/Wed. If it was starting on a Thursday, it jumps to the next Monday automatically.

TASK 5.3: Task Types — Fixed Units vs Fixed Duration vs Fixed Work

The scheduling formula is: $\text{Work} = \text{Duration} \times \text{Units}$. A task type controls WHICH of the 3 values stays fixed when the other two change.

Task Type	Fixed Value	You Change	Project Recalculates
Fixed Units (default)	Units (%)	Duration	Work recalculates
Fixed Units (default)	Units (%)	Work	Duration recalculates
Fixed Duration	Duration	Units	Work recalculates
Fixed Duration	Duration	Work	Peak Units recalculates
Fixed Work	Work	Duration	Peak Units recalculates
Fixed Work	Work	Units	Duration recalculates

Exercise A — Fixed Units (Default): Change Duration → Work Recalculates

Task: System Integration Testing | Type: Fixed Units (default) | Currently: 4 days, Dev1 @ 100% = 32 hrs

Step 1: Go to View → Task Views → Task Usage (to see Work values clearly).

Step 2: Find 'System Integration Testing'. Note Duration = 4d, Work = 32 hrs.

Step 3: Change duration to 6d (type in Duration field, press Enter).

MS Project increases Work from 32 → 48 hrs. Units stay at 100%.

Step 4: Click the Action button (small green triangle on task name). Review options.

Step 5: Click 'Decrease the hours resources work per day (units) but keep the same amount of work' if you want to keep 32 hrs and reduce units instead.

✓ **RESULT:** Fixed Units: duration changed → work recalculated. Action button lets you override.

Report Overview:

PROJECT OVERVIEW

MON 06/01/25 MON 31/03/25

PROJECT OVERVIEW



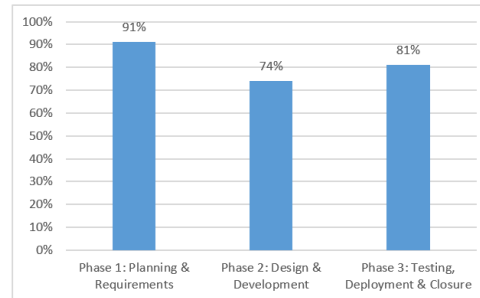
MILESTONES DUE

Milestones that are coming soon.

Name	Finish
M1: Phase 1 Complete	Fri 24/01/25
M2: Development Complete	Mon 10/03/25

% COMPLETE

Status for all top-level tasks. To see the status for subtasks, click on the chart and update the outline level in the Field List.



LATE TASKS

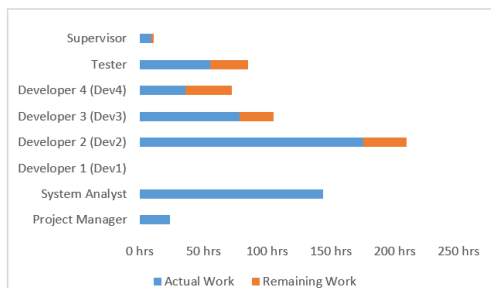
Tasks that are past due.

Work Overview:

Actual Work Remaining Work Baseline Work

WORK STATS

Shows work stats for all top level tasks.



WORK OVERVIEW

Mon 06/01/25 - Mon 31/03/25

